JOINT UNIVERSITIES PRELIMINARY EXAMINATIONS BOARD

SEPTEMBER 2021 EXAMINATIONS

CHEMISTRY PRACTICAL **Time Allowed: 2½ Hours**

Answer All Questions.

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QUESTION 1: QUANTITATIVE ANALYSIS

You are provided with the following;

A is solution of HCl containing 0.25 mole in 500 cm^3 of solution. **B** is a solution containing a mixture of $NaHCO_3$ and Na_2CO_3.

(a) Put **A** into the burette and titrate it against 25 cm^3 portion of B using 2-3 drops of phenolphthalein indicator. Stop titration just at the point when there is a slight permanent colour change of phenolphthalein indicator used, and record the volume of **A** used with this indicator. Then add 2-3 drops of methyl orange indicator to the same titration mixture in the conical flask and continue your titration until the end point is reached.

Record the volume of **A** used at the end point of methyl orange indicator. Repeat the same titration procedure 3 times.

Determine:

(i) Average volume of acid used for neutralizing Na_2CO_3,

(ii) Average volume of acid A used with $NaHCO_3$.

Equations involved in the reaction are:

$$Na_2CO_3\,(aq) + HCl\,(aq) \rightarrow NaHCO_3\,(aq) + NaCl\,(aq)$$

$$NaHCO_3\,(aq) + HCl\,(aq) \rightarrow NaCl\,(aq) + CO_2\,(g) + H_2O\,(l)$$

(b) From the information provided in (a) and your results, calculate the:

(i) Concentration of Na_2CO_3 in moles/dm^3 in solution **B**

(ii) Concentration of Na_2CO_3 in g/dm^3 in solution **B**

(iii) Mole of $NaHCO_3$ in 25 cm^3 of solution used.

(iv) Amount in mole of $NaHCO_3$ in 1dm^3 of solution **B**

(v) Amount in gram of $NaHCO_3$ in 1dm^3 of solution **B**

(vi) Percentage of Na_2CO_3 in 1dm^3 of solution B, if 1 dm^3 of solution B contains 12 g of a mixture of Na_2CO_3 and $NaHCO_3$. [Na=23, C=12, O=16, H=1]

[Total = 10 marks]

QUESTION 2: QUALITATIVE ANALYSIS

You are provided with sample **C**, a mixture of 3 simple salts and are requested to perform the following experiments to identify the acidic radicals. Follow the instructions provided. Present your report in a tabular form with the headings, Test, Observation and Inference.

Ensure your test tube is clean and dry.

Test 1: Introduce **C** into the test tube provided, add 2 M HCl, test the gas evolved with lime water

Test 2: Introduce **C** into the test tube provided, add conc H_2SO_4, test gas evolved with conc. Ammonia under the fume hood. Thereafter, add minimum MnO_2 and heat.

Test 3: Introduce **C** into the test tube provided, add conc H_2SO_4, introduce copper tunings' and heat.

For each test carried out, give a brief description of the chemical reactions involved showing equations where relevant.

[Total = 10 marks]